Questions by topic: Soils: fertility, erosion & management - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the textural composition of four soil samples.

Table 1

Soil sample	Sand / %	Silt / %	Clay / %
A	85	10	5
B	40	40	20
C	20	25	55
D	10	80	10

0 1 . 1 Identify the sample that is a loam and explain why loams are ideal for cultivating most crops.

[3 marks]

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0 1 . 2 Explain why sample A is likely to have the highest permeability but the lowest nutrient retention.

[3 marks]

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0 1 . 3 Compare the thermal capacity and root penetration of soils A and C.

[2 marks]

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0 1 . 4 Describe how the texture of a soil sample could be determined by sedimentation.

[2 marks]

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0 2 . 1 Explain the roles of **three** named groups of soil organisms in maintaining soil fertility.

[6 marks]

0 2 . 2 Explain why a soil pH between 5.5 and 7.0 is best for fertility.

[2 marks]

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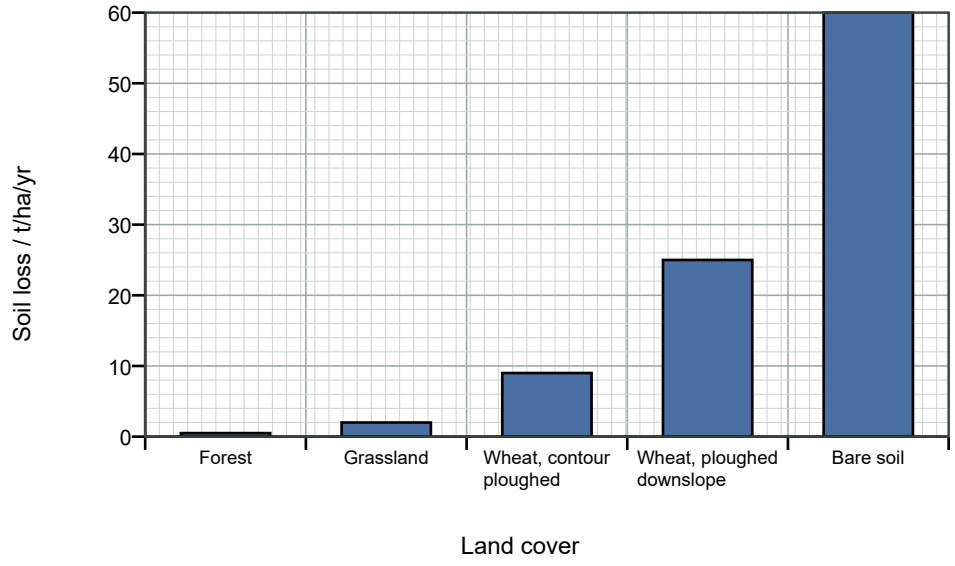
0 2 . 3 Explain how soil compaction by heavy machinery reduces fertility.

[2 marks]

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0 3 Figure 1 shows the annual soil loss measured on experimental plots with different vegetation cover on a 10% slope in a region with heavy seasonal rain.

Figure 1



0 3 . 1 Describe the pattern shown in Figure 1.

[2 marks]

0 3 . 2 Explain **four** ways in which vegetation reduces the rate of soil erosion.

[4 marks]

0 3 . 3 Explain why ploughing up and down a slope causes more erosion than contour ploughing.

[2 marks]

0 3 . 4 Explain why the region's climate makes the soil vulnerable to **both** wind and water erosion.

[2 marks]

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0 4 . 1 Explain how each of the following human activities increases soil erosion.

[4 marks]

Overgrazing .....

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Reducing soil biota .....

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0 4 . 2 Explain why landslides become more likely on steep slopes some years after deforestation.

[2 marks]

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0 4 . 3 Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.

[4 marks]

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0 5 The Universal Soil Loss Equation (USLE) is A = R x K x LS x C x P. Table 1 gives factor values for a farm.

Table 1

Factor	Description	Value
R	Rainfall erosivity for the farm	110
K	Soil erodibility: loam	0.67
LS	Slope length and gradient: 120 m, 6%	1.35
C (crop type x tillage)	Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)	see question
P	Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25	see question

Table 2

Soil erosion class	Rate of soil loss / t ha^{-1} yr^{-1}
Low	under 5
Moderate	5-10
High	10-20
Very high	20-30
Severe	over 30

0 5 . 1 Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

[2 marks]

Answer _____ t/ha/yr

0 5 . 2 Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

[2 marks]

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Answer [ ] t/ha/yr

0 5 . 3 The natural rate of soil formation on the farm is 1 t ha^{-1} yr^{-1}. Use your answers and Table 2 to evaluate the sustainability of the two farming systems.

[3 marks]

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0 5 . 4 Explain why the USLE gives only an estimate of soil loss.

[2 marks]

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0 5 . 5 Suggest which single factor in the USLE a farmer can change most easily.

[1 mark]

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0 6 . 1 Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]

Table 1

Method	How it reduces erosion
Terracing	
Tied ridging	
Windbreaks / hedgerows	
Mulching	
Strip cropping	

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0 6 . 2 Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

0 6 . 3 Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

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0 7 A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

0 7 . 1 Calculate the percentage water content of the fresh soil.

[2 marks]

Answer ..... %

0 7 . 2 Calculate the percentage organic matter content of the dry soil.

[2 marks]

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Answer [ ] %

0 7 . 3 Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

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0 7 . 4 Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

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0 7 . 5 Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

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0 8 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 8 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 8 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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END OF QUESTIONS

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